

SG/PG Series

GENERAC®

**INDUSTRIAL
POWER**

350 kW

21.9L

Industrial Spark-Ignited Generator Set

Generac International Products

60 Hz



*Built in the USA using domestic and foreign parts

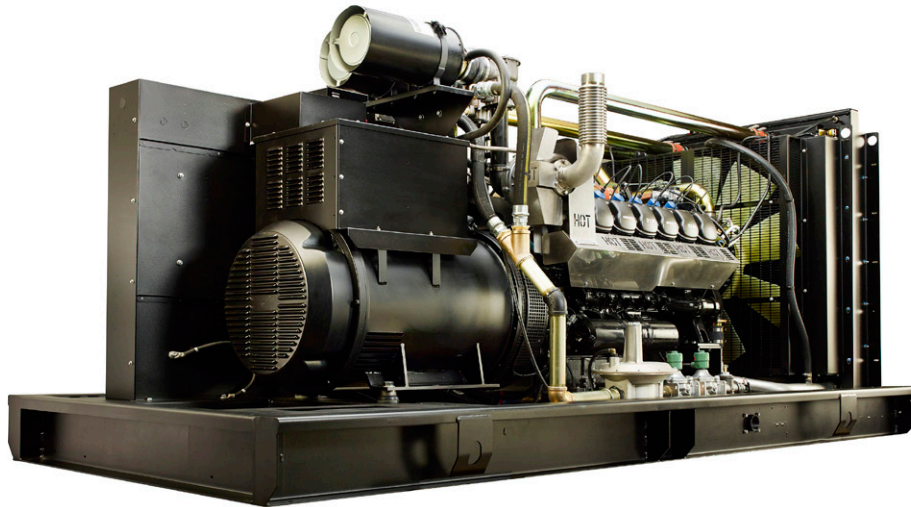


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Power Ratings		
Standby	SG350	350 kW
Prime	PG315	315 kW

Powering Ahead

For over 50 years, Generac has provided innovative design and superior manufacturing.

Generac ensures superior quality by designing and manufacturing most of its generator components, including alternators, enclosures and base tanks, control systems and communications software.

Generac's gensets utilize a wide variety of options, configurations and arrangements, allowing us to meet the standby power needs of practically every application.

Generac searched globally to ensure the most reliable engines power our generators. We choose only engines that have already been proven in heavy-duty industrial applications under adverse conditions.

Generac is committed to ensuring our customers' service support continues after their generator purchase.

SG/PG Series

Standard Features

ENGINE SYSTEM

General

- Oil Drain Extension
- Air Cleaner
- Fan Guard
- Stainless Steel flexible exhaust connection
- Critical Exhaust Silencer
- Factory Filled Oil
- Radiator duct adapter (open set only)

Fuel System

- Primary and Secondary Fuel Shutoff

Cooling System

- Closed Coolant Recovery System
- UV/Ozone resistant hoses
- Factory-installed Radiator
- Radiator drain extension
- 50/50 Ethylene glycol antifreeze

Engine Electrical System

- Battery charging alternator
- Battery Cables
- Battery Tray
- Solenoid activated starter motor
- Rubber-booted engine electrical connections

ALTERNATOR SYSTEM

- UL2200 GENprotect™
- Class H insulation material
- 2/3 Pitch
- Skewed Stator
- Permanent Magnet Excitation
- Sealed Bearings
- Amortisseur winding
- Full load capacity alternator

GENERATOR SET

- Internal Genset Vibration Isolation
- Separation of circuits - high/low voltage
- Separation of circuits - multiple breakers
- Wrapped Exhaust Piping (enclosed set only)
- Standard Factory Testing
- 2 Year Limited Warranty (Standby rated Units)
- 1 Year Warranty (Prime rated units)
- Silencer mounted in the discharge hood (enclosed only)

ENCLOSURE (if selected)

- Rust-proof fasteners with nylon washers to protect finish
- High performance sound-absorbing material
- Gasketed doors
- Stamped air-intake louvers
- Air discharge hoods for radiator-upward pointing
- Stainless steel lift off door hinges
- Stainless steel lockable handles
- Rhino Coat™ - Textured polyester powder coat

CONTROL SYSTEM



Control Panel

- Digital H Control Panel - Dual 4x20 Display
- Programmable Crank Limiter
- 7-Day Programmable Exerciser
- Special Applications Programmable PLC
- RS-232/485
- All-Phase Sensing DVR
- Full System Status
- Utility Monitoring
- Low Fuel Pressure Indication
- 2-Wire Start Compatible
- Power Output (kW)
- Power Factor
- kW Hours, Total & Last Run

- Real/Reactive/Apparent Power
- All Phase AC Voltage
- All Phase Currents
- Oil Pressure
- Coolant Temperature
- Coolant Level
- Engine Speed
- Battery Voltage
- Frequency
- Date/Time Fault History (Event Log)
- Isochronous Governor Control
- Waterproof/sealed Connectors
- Audible Alarms and Shutdowns
- Not in Auto (Flashing Light)
- Auto/Off/Manual Switch
- E-Stop (Red Mushroom-Type)
- NFPA110 Level I and II (Programmable)
- Customizable Alarms, Warnings, and Events
- Modbus protocol
- Predictive Maintenance algorithm
- Sealed Boards
- Password parameter adjustment protection

- Single point ground
- 15 channel data logging
- 0.2 msec high speed data logging
- Alarm information automatically comes up on the display

Alarms

- Oil Pressure (Pre-programmable Low Pressure Shutdown)
- Coolant Temperature (Pre-programmed High Temp Shutdown)
- Coolant Level (Pre-programmed Low Level Shutdown)
- Low Fuel Pressure Alarm
- Engine Speed (Pre-programmed Over speed Shutdown)
- Battery Voltage Warning
- Alarms & warnings time and date stamped
- Alarms & warnings for transient and steady state conditions
- Snap shots of key operation parameters during alarms & warnings
- Alarms and warnings spelled out (no alarm codes)

SG/PG Series

Configurable Options

ENGINE SYSTEM

- General
- Engine Block Heater with ball valves
- Flexible Fuel Line - NPT Connection
- Oil Heater
- Air Filter Restriction Indicator
- Stone Guard (open set only)

Engine Electrical System

- 10A UL battery charger
- Battery Warmer

ALTERNATOR SYSTEM

- Alternator Upsizing
- Anti-Condensation Heater
- Tropical coating (480/600 V non-upsized only)

GENERATOR SET

- Gen-Link Communications Software (English Only)
- Extended Factory Testing (3 Phase Only)
- 8 Position Load Center

CIRCUIT BREAKER OPTIONS

- Main Line Circuit Breaker
- 2nd Main Line Circuit Breaker
- Shunt Trip and Auxiliary Contact
- Electronic Trip Breakers

ENCLOSURE

- Standard Enclosure
- Level 1 Sound Attenuation
- Level 2 Sound Attenuation
- Steel Enclosure
- Aluminum Enclosure
- 180 MPH Wind Kit
- 12 VDC Enclosure Lighting Kit
- 120 VAC Enclosure Lighting Kit
- AC/DC Enclosure Lighting Kit

CONTROL SYSTEM

- | | | |
|---|---|--|
| <ul style="list-style-type: none"> ○ 21-Light Remote Annunciator ○ Remote Relay Panel (8 or 16) ○ Oil Temperature Sender with Indication Alarm | <ul style="list-style-type: none"> ○ Remote E-Stop (Break Glass-Type, Surface Mount) ○ Remote E-Stop (Red Mushroom-Type, Surface Mount) ○ Remote E-Stop (Red Mushroom-Type, Flush Mount) | <ul style="list-style-type: none"> ○ Remote Communication - Modem ○ Remote Communication - Ethernet ○ 10A Run Relay ○ Ground fault indication and protection functions |
|---|---|--|

Engineered Options

ENGINE SYSTEM

- Coolant heater ball valves
- Fluid containment pans
- Low fuel pressure system (7"-11" H₂O)

ALTERNATOR SYSTEM

- 3rd Breaker Systems

GENERATOR SET

- Special Testing
- Battery Box

ENCLOSURE

- Motorized Dampers
- Enclosure Ambient Heaters
- Door Alarm Switch

CONTROL SYSTEM

- Spare inputs (x4) / outputs (x4) - H Panel Only
- Battery Disconnect Switch

Rating Definitions

Standby – Applicable for a varying emergency load for the duration of a utility power outage with no overload capability.

Prime – Applicable for supplying power to a varying load in lieu of utility for an unlimited amount of running time. A 10% overload capacity is available for 1 out of every 12 hours. The Prime Power option is only available on International applications.

Power ratings in accordance with ISO 8528-1, Second Edition dated 2005-06-01, definitions for Prime Power (PRP) and Emergency Standby Power (ESP).

SG/PG Series

application and engineering data

ENGINE SPECIFICATIONS

General

Make	Generac
Cylinder #	12
Type	V12
Displacement - L (Cu In)	21.9 (1336.42)
Bore - mm (in)	128 (5.03)
Stroke - mm (in)	142 (5.6)
Compression Ratio	10:1
Intake Air Method	Turbocharged/Aftercooled
Number of Main Bearings	7
Connecting Rods	Alloy Steel
Cylinder Head	Cast Iron - OHV
Cylinder Liners	Cast Alloy Steel
Ignition	Altronic CD200D
Pistons	Aluminum Alloy
Crankshaft	Forged Alloy Steel
Lifter Type	Solid
Intake Valve Material	High Temp Alloy Steel
Exhaust Valve Material	High Temp Alloy Steel
Hardened Valve Seats	High Temp Alloy Steel

Engine Governing

Governor	Electronic
Frequency Regulation (Steady State)	(+/-) 0.25%

Lubrication System

Oil Pump Type	Gear
Oil Filter Type	Twin Full flow with intercooler
Crankcase Capacity - L (qts)	30 (31.7)

Cooling System

Cooling System Type	Pressurized Closed Recovery
Water Pump Flow - gpm (lpm)	211 (800)
Fan Type	Pusher
Fan Speed (rpm)	1404
Fan Diameter mm (in)	44
Coolant Heater Wattage	2500
Coolant Heater Standard Voltage	240 V

Fuel System

Fuel Type	Natural Gas
Carburetor	Down Draft
Secondary Fuel Regulator	Standard
Fuel Shut Off Solenoid	Standard (Dual)
Operating Fuel Pressure	11" - 15" H ₂ O
Operating Fuel Pressure (Optional)	7" - 11" H ₂ O

Engine Electrical System

System Voltage	24 VDC
Battery Charging Alternator	Std
Battery Size	See Battery Index 0161970SBY
Battery Voltage	(2) 12 VDC
Ground Polarity	Negative

ALTERNATOR SPECIFICATIONS

Standard Model	520
Poles	4
Field Type	Revolving
Insulation Class - Rotor	H
Insulation Class - Stator	H
Total Harmonic Distortion	<5%
Telephone Interference Factor (TIF)	<50
Standard Excitation	Permanent Magnet
Bearings	Sealed Ball
Coupling	Direct, Flexible Disc
Prototype Short Circuit Test	Yes

Voltage Regulator Type	Full Digital
Number of Sensed Phases	5
Regulation Accuracy (Steady State)	+/- 0.25%

SG/PG Series

operating data

60 Hz

POWER RATINGS – NATURAL GAS

	Standby		Prime	
Three-Phase 120/208 VAC @0.8pf	350 kW	1216 Amps	315 kW	TBD Amps
Three-Phase 120/240 VAC @0.8pf	350 kW	1053 Amps	315 kW	TBD Amps
Three-Phase 277/480 VAC @0.8pf	350 kW	527 Amps	315 kW	TBD Amps
Three-Phase 346/600 VAC @0.8pf	350 kW	421 Amps	315 kW	TBD Amps

STARTING CAPABILITIES (sKVA)

		sKVA vs. Voltage Dip											
		480 VAC						208/240 VAC					
Alternator	kW	10%	15%	20%	25%	30%	35%	10%	15%	20%	25%	30%	35%
Standard	350	387	581	775	968	1162	1356	345	570	835	1100	1450	1710
Upsize 1	555	457	686	914	1143	1371	1600	-	-	-	-	-	-
Upsize 2	642	471	707	943	1179	1414	1650	543	814	1086	1357	1629	1900

FUEL CONSUMPTION RATES*

Natural Gas – ft³/hr (m³/hr)		
Percent Load	Standby	Prime
25%	1732 (49)	1559 (44.1)
50%	2598 (73.6)	2338 (66.2)
75%	3463 (98.1)	3117 (88.3)
100%	4328 (122.6)	3895 (110.3)

*Fuel supply installation must accommodate fuel consumption rates at 100% load.

COOLING

		Standby	Prime
Cooling Fan Air Flow (Open Set)	cfm (m³/min)	25,100 (711)	25,100 (711)
Coolant Flow per Minute	gpm (lpm)	211 (800)	211 (800)
Coolant System Capacity	gal (L)	23 (87)	23 (87)
Heat Rejection to Coolant	BTU/hr	1,194,528	1,099,032
Max. Ambient Temp (before derate)†	°F (°C)	104 (40)	104 (40)
Maximum External Static Resistance	in H ₂ O	0.5	0.5

COMBUSTION AIR REQUIREMENTS

		Standby	Prime
Flow at Rated Power	cfm (m³/min)	1019 (29)	974 (28)

ENGINE

		Standby	Prime
Rated Engine Speed	rpm	1800	1800
Engine Horsepower**	hp	620	620
Piston Speed	ft/min (m/min)	-	-
BMEP	psi	185	168

** Refer to "Emissions Data Sheet" for maximum bHP for EPA and SCAQMD permitting purposes.

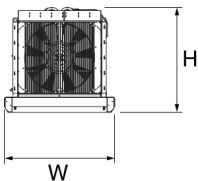
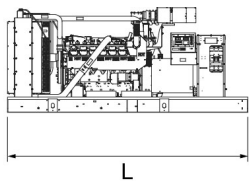
EXHAUST

		Standby	Prime
Total Exhaust Flow (Rated Output)	cfm (m³/min)	3626 (103)	3428 (97)
Maximum Additional Backpressure (Post Silencer)	inHg	0.75	0.75
Exhaust Temp (Rated Output - post silencer)	°F (°C)	1350 (732)	1350 (732)
Exhaust Outlet Size (Open Set)	in	3.5" O.D. Flex	3.5" O.D. Flex

† Deration – Operational characteristics consider maximum ambient conditions. Derate factors may apply under atypical site conditions. Please consult a Generac Power Systems Industrial Dealer for additional details. All performance ratings in accordance with ISO3046, BS5514, ISO8528 and DIN6271 standards.

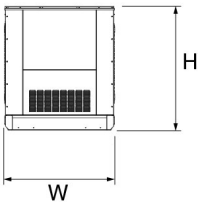
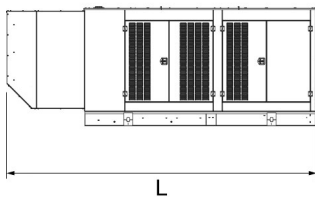
SG/PG Series

dimensions, weights, and sound levels



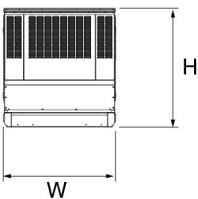
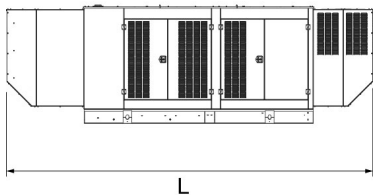
OPEN SET (Includes Exhaust Flex)

L x W x H in (mm)	154.4 (3923) x 71 (1803) x 67 (1702)
Weight lbs (kg)	8429 (3823)
Sound Level (dBA*)	93



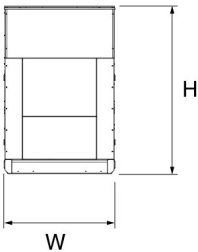
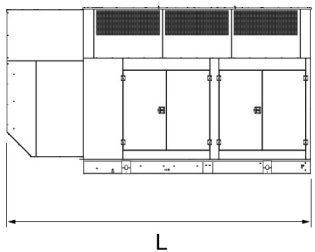
STANDARD ENCLOSURE

L x W x H in (mm)	207.4 (5268) x 71 (1803) x 80 (2032)
Weight lbs (kg)	Steel: 10428 (4730) Aluminum: 9298 (4217)
Sound Level (dBA*)	92



LEVEL 1 ACOUSTIC ENCLOSURE

L x W x H in (mm)	247.5 (6285) x 71 (1803) x 80 (2032)
Weight lbs (kg)	Steel: 11211 (5085) Aluminum: 9720 (4409)
Sound Level (dBA*)	82



LEVEL 2 ACOUSTIC ENCLOSURE

L x W x H in (mm)	207.4 (5268) x 71 (1803) x 114 (2899)
Weight lbs (kg)	Steel: 11759 (5333) Aluminum: 9951 (4513)
Sound Level (dBA*)	75

*All measurements are approximate and for estimation purposes only. Sound levels measured at 23 ft (7 m) and does not account for ambient site conditions.

YOUR FACTORY RECOGNIZED GENERAC INDUSTRIAL DEALER

Specification characteristics may change without notice. Please consult a Generac Power Systems Industrial Dealer for detailed installation drawings.